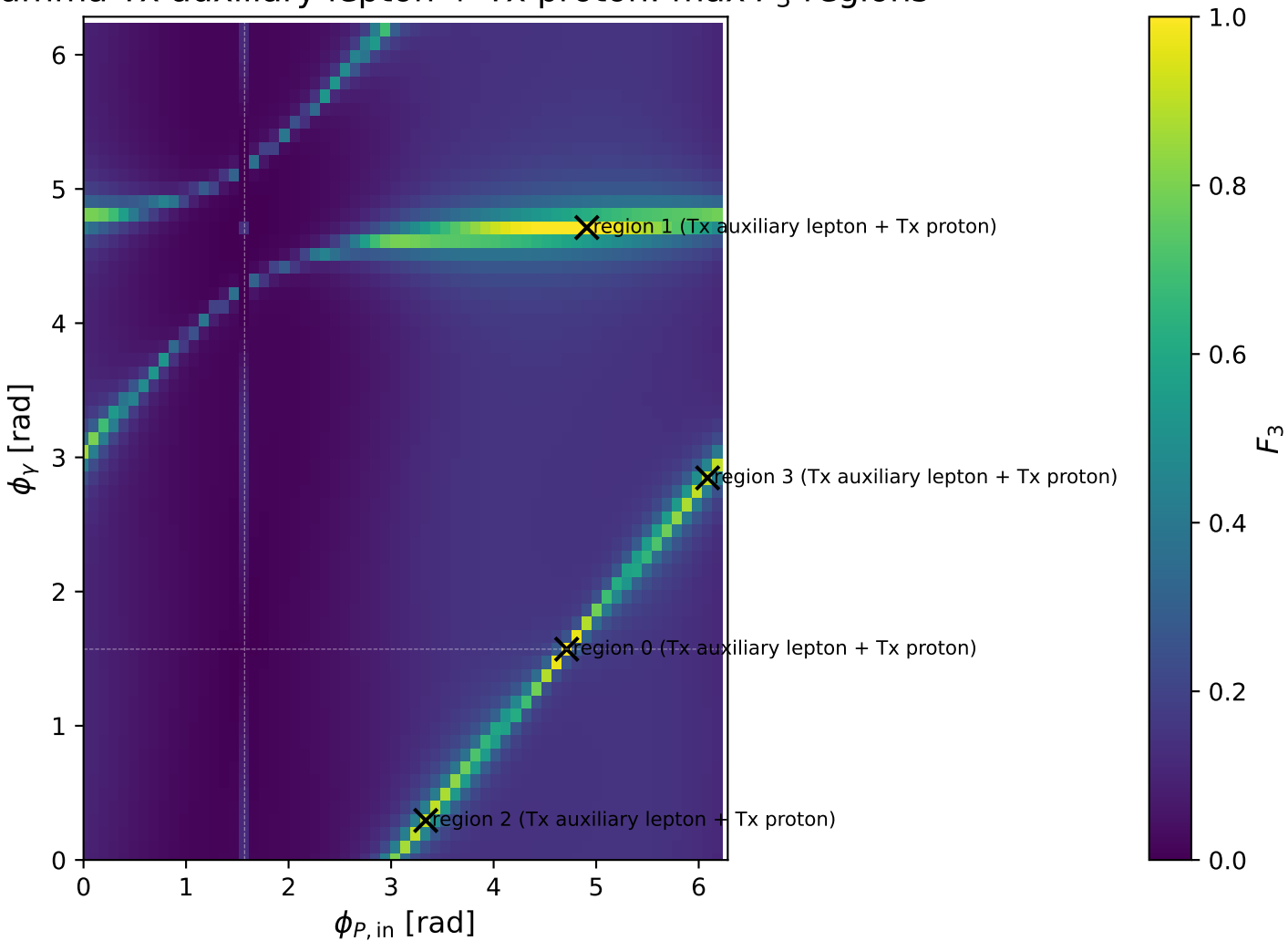
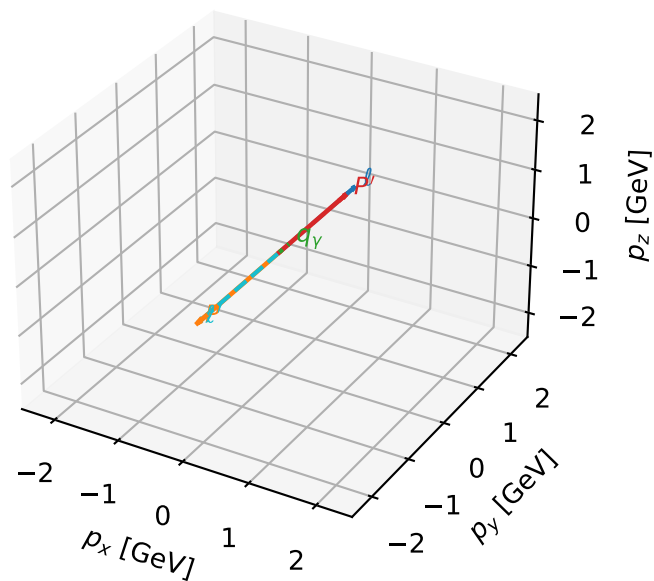


gamma Tx auxiliary lepton + Tx proton: max F_3 regions



Tx auxiliary lepton + Tx proton characteristic max F_3 configuration

3D momenta

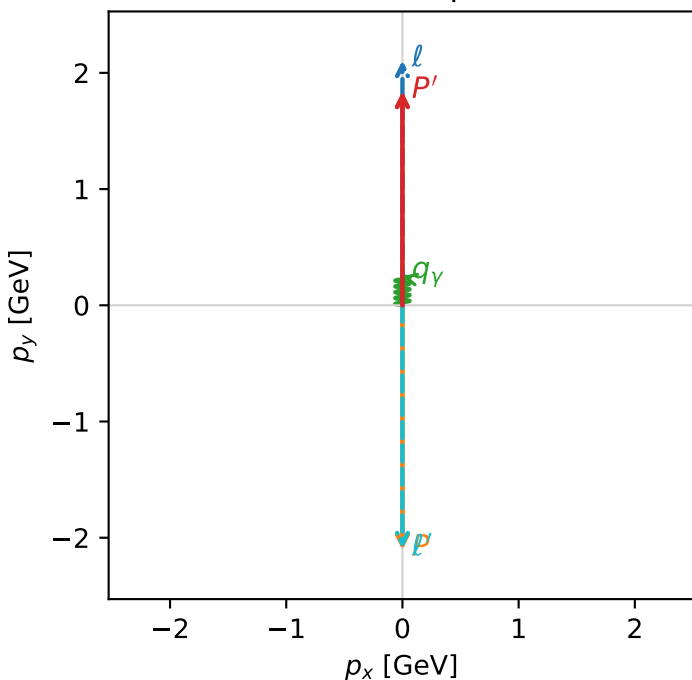


f3_low_egamma_txtx_region_0 F_3=0.996521
 region: low_Egamma_TxTx_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\ell}\rangle \otimes |\bar{T}_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.84352$
 $\theta_{in}=1.57079$, $\phi_\ell=1.5708$, $\phi_p=4.71239$
 $E_\gamma=0.25$, $\phi_\gamma=1.5708$
 $Q^2=17.6437$, $x_B=0.993667$, $t=-15.5495$, $W^2=0.9923$, $y=0.904266$
 $\theta(\ell', \gamma)=180$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= 0.0000$
 ℓ' $E= 2.3201$ $p_x= -0.0000$ $p_y= -2.0935$ $p_z= 0.0000$
 P' $E= 2.0684$ $p_x= 0.0000$ $p_y= 1.8435$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.0000$ $p_y= 0.2500$ $p_z= 0.0000$

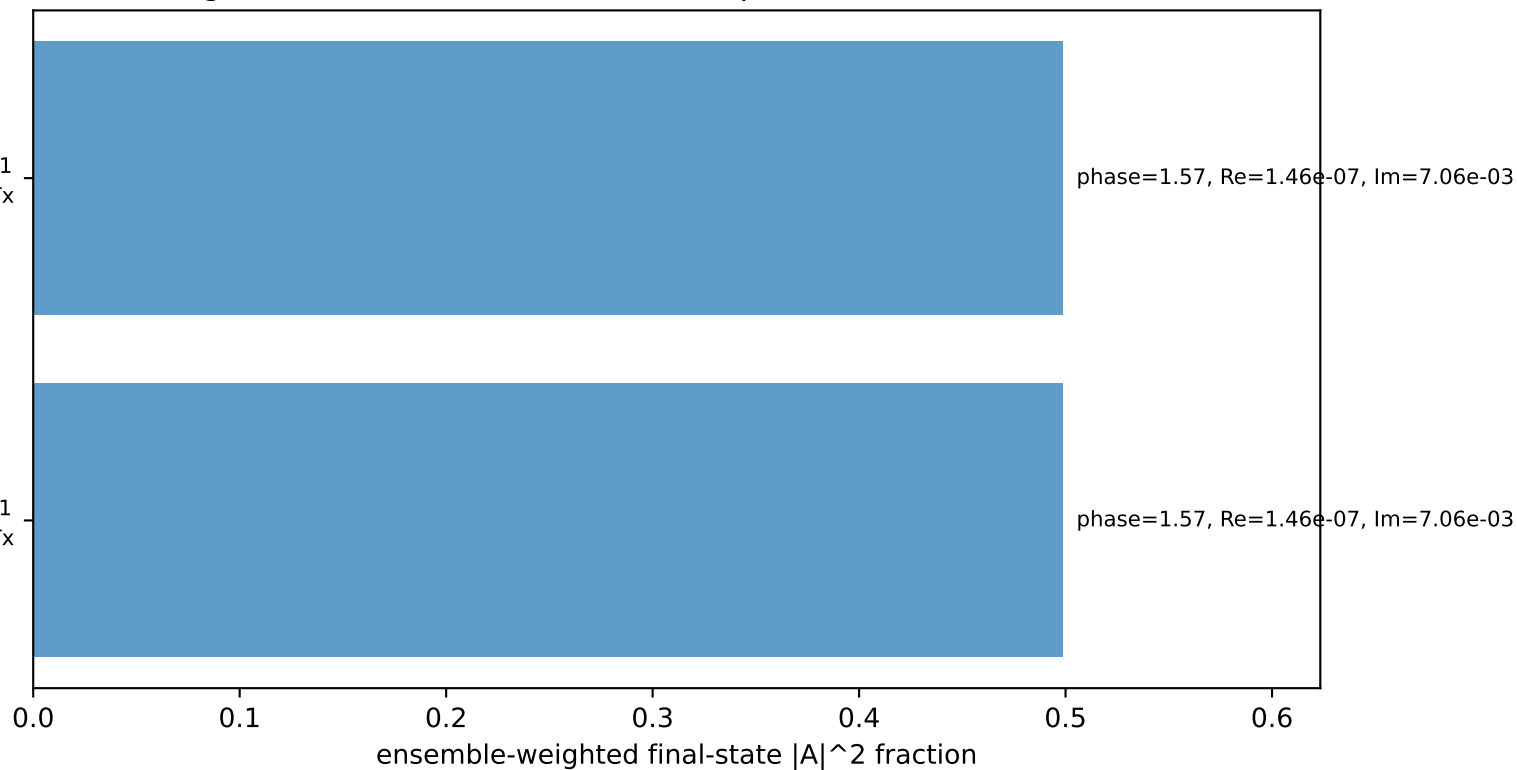
Transverse plane



$h_e=-1$, $h_p=-1$, $h_\gamma=+1$
 auxiliary lepton Tx, proton Tx

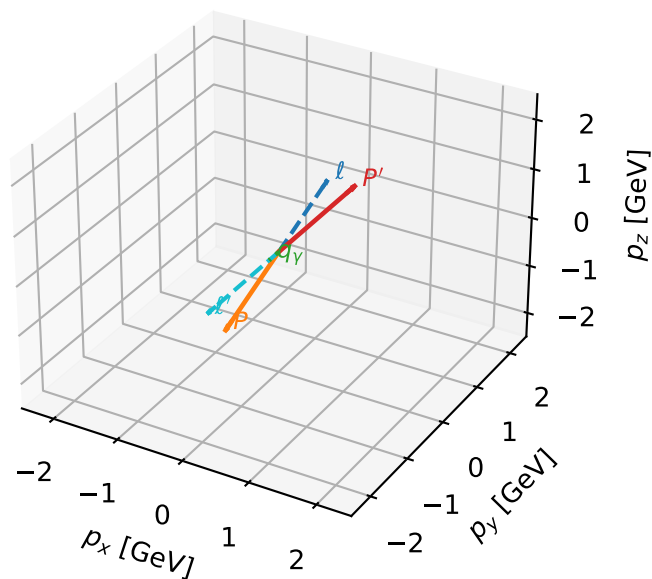
$h_e=+1$, $h_p=+1$, $h_\gamma=-1$
 auxiliary lepton Tx, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=99.7%)



Tx auxiliary lepton + Tx proton characteristic max F_3 configuration

3D momenta

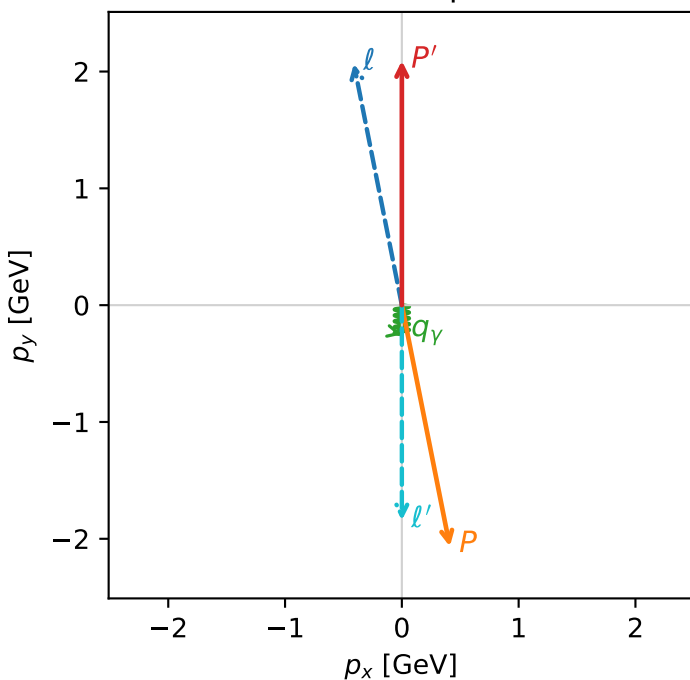


f3_low_egamma_txtx_region_1 $F_3=0.996033$
 region: low_Egamma_TxTx_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\ell}\rangle \otimes |\bar{T}_{x,p}\rangle$

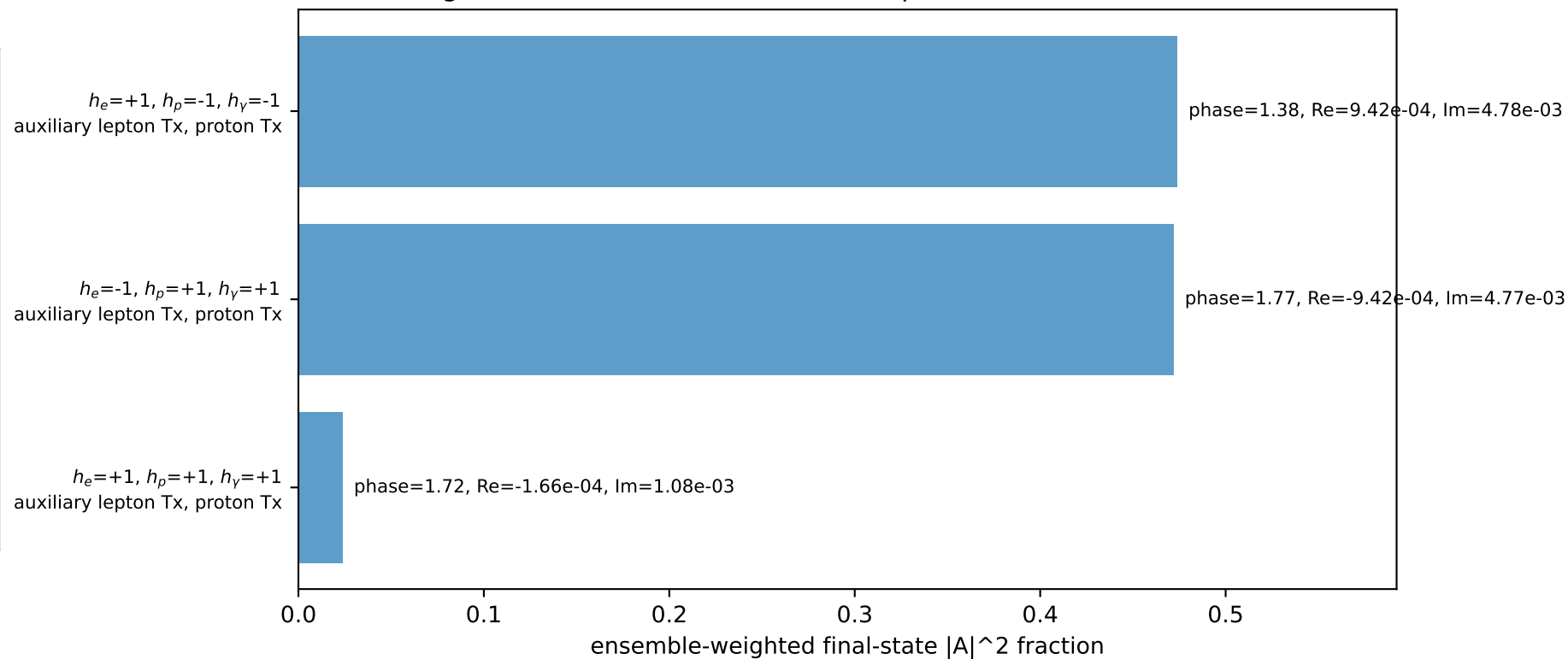
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=2.09195$
 $\theta_{in}=1.57079$, $\phi_\ell=1.76715$, $\phi_p=4.90874$
 $E_\gamma=0.25$, $\phi_\gamma=4.71239$
 $Q^2=15.3888$, $x_B=0.875304$, $t=-17.4611$, $W^2=3.07213$, $y=0.895349$
 $\theta(\ell', \gamma)=0$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -0.4110$ $p_y= 2.0665$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 0.4110$ $p_y= -2.0665$ $p_z= 0.0000$
 ℓ' $E= 2.0959$ $p_x= 0.0000$ $p_y= -1.8420$ $p_z= 0.0000$
 P' $E= 2.2926$ $p_x= 0.0000$ $p_y= 2.0920$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.0000$ $p_y= -0.2500$ $p_z= 0.0000$

Transverse plane

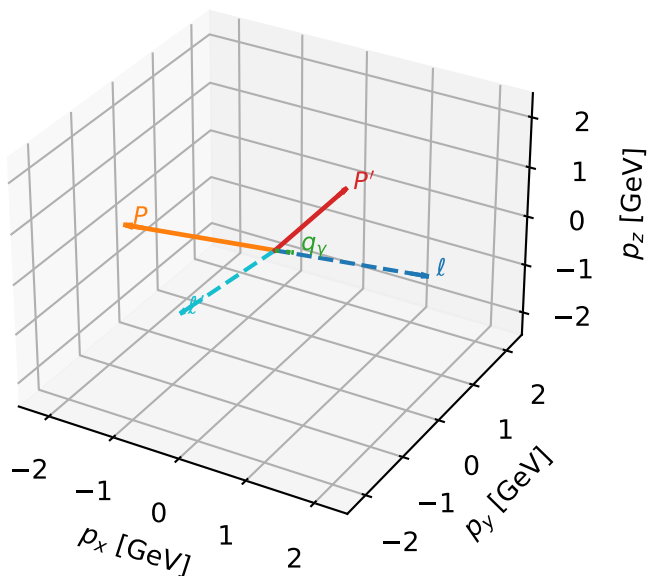


Leading initial-ensemble/final-state components (N=3, retained=96.9%)



Tx auxiliary lepton + Tx proton characteristic max F_3 configuration

3D momenta

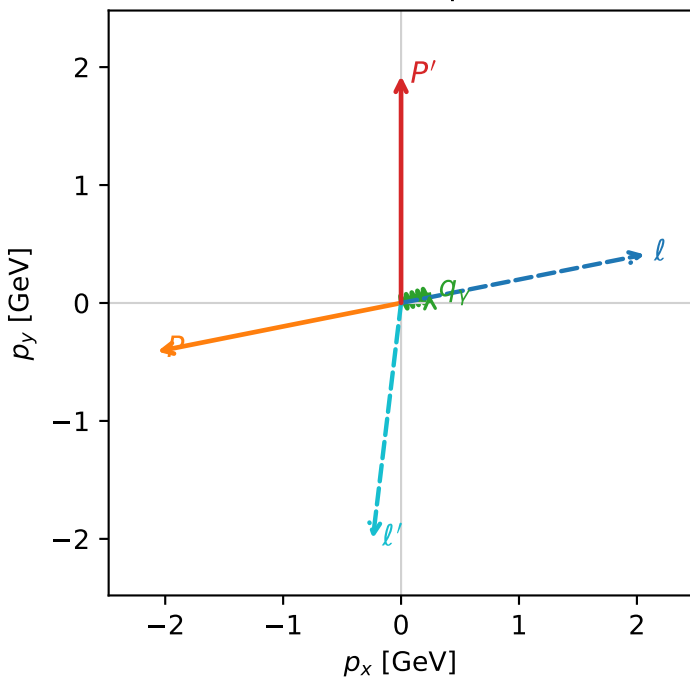


f3_low_egamma_tctx_region_2 $F_3=0.91574$
 region: low_Egamma_TxTx_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\ell}\rangle \otimes |\bar{T}_{x,p}\rangle$

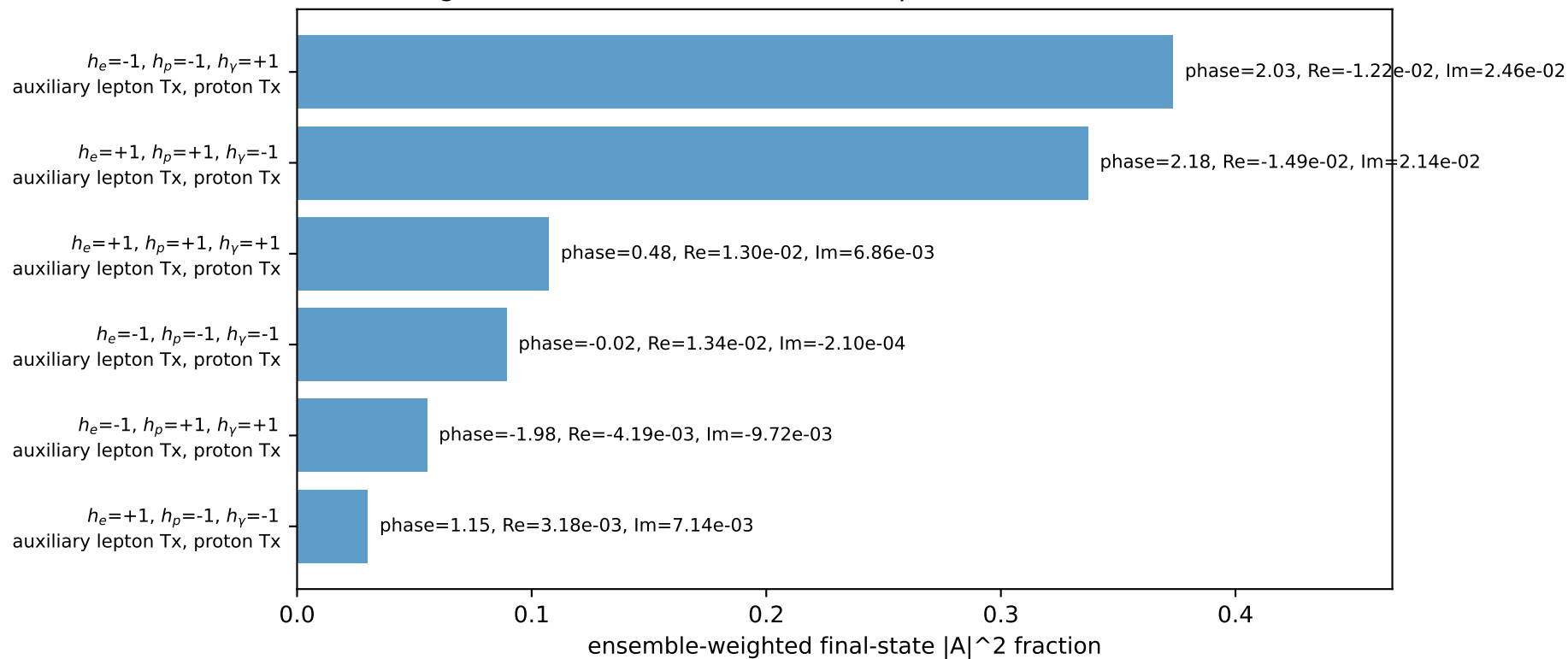
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.92527$
 $\theta_{in}=1.57079$, $\phi_\ell=0.19635$, $\phi_P=3.33794$
 $E_\gamma=0.25$, $\phi_\gamma=0.294524$
 $Q^2=11.1117$, $x_B=0.933515$, $t=-9.70147$, $W^2=1.67121$, $y=0.606184$
 $\theta(\ell', \gamma)=113.703$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 2.0665$ $p_y= 0.4110$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -2.0665$ $p_y= -0.4110$ $p_z= 0.0000$
 ℓ' $E= 2.2469$ $p_x= -0.2392$ $p_y= -1.9978$ $p_z= 0.0000$
 P' $E= 2.1416$ $p_x= 0.0000$ $p_y= 1.9253$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2392$ $p_y= 0.0726$ $p_z= 0.0000$

Transverse plane

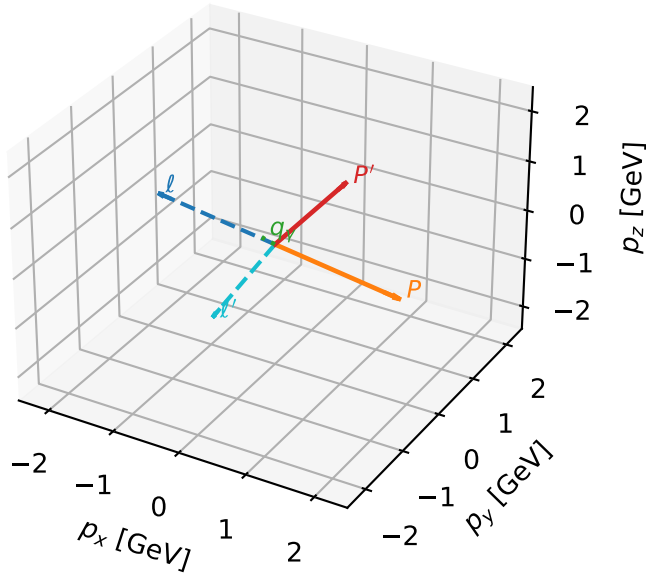


Leading initial-ensemble/final-state components (N=6, retained=99.3%)



Tx auxiliary lepton + Tx proton characteristic max F_3 configuration

3D momenta

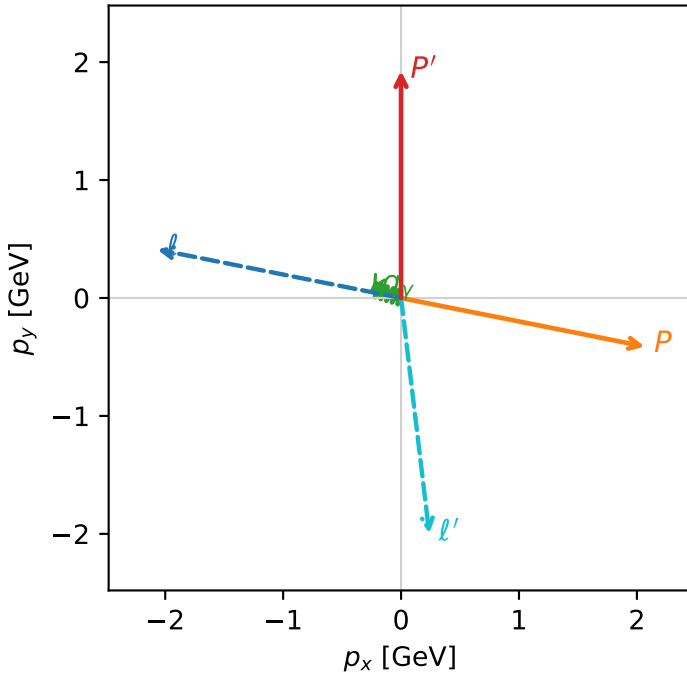


f3_low_egamma_txtx_region_3 $F_3=0.91574$
 region: low_Egamma_TxTx_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,l}\rangle \otimes |\bar{T}_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.92527$
 $\theta_{in}=1.57079$, $\phi_l=2.94524$, $\phi_P=6.08684$
 $E_\gamma=0.25$, $\phi_\gamma=2.84707$
 $Q^2=11.1117$, $x_B=0.933515$, $t=-9.70147$, $W^2=1.67121$, $y=0.606184$
 $\theta(l', \gamma)=113.703$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= -2.0665$ $p_y= 0.4110$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 2.0665$ $p_y= -0.4110$ $p_z= 0.0000$
 l' $E= 2.2469$ $p_x= 0.2392$ $p_y= -1.9978$ $p_z= 0.0000$
 P' $E= 2.1416$ $p_x= 0.0000$ $p_y= 1.9253$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.2392$ $p_y= 0.0726$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=6, retained=99.3%)

